

DRY-RUN GUIDE | sum.py

Source: Python\1st_Belt\sum.py

CORE BEHAVIOR

Reads one non-negative integer and adds its decimal digits by repeatedly taking '% 10' and integer-dividing by 10.

REPRESENTATIVE INPUT

'5024'

STEP TRACE

Start total=0. Digit 4 -> 4; 2 -> 6; 0 -> 6; 5 -> 11. num becomes 0 and the loop ends.

EXPECTED OUTPUT / RETURN

'11'

ACTUAL SOURCE EXCERPT

Verbatim excerpt from Python\1st_Belt\sum.py; line numbers refer to the original file.
No source was modified.

```
001 | # Question: Calculate the sum of the given numbers.
002 | # Add your solution here.
003 |
004 | num=int(input("Enter a non-negative integer: "))
005 | total = 0
006 | while num > 0:
007 |     digit = num % 10
008 |     total += digit
009 |     num //= 10
010 | print(total)
```

DRY-RUN NOTE

Values are representative. For function-only modules the guide uses an illustrative direct call and explains its return; quirks/errors describe actual behavior.